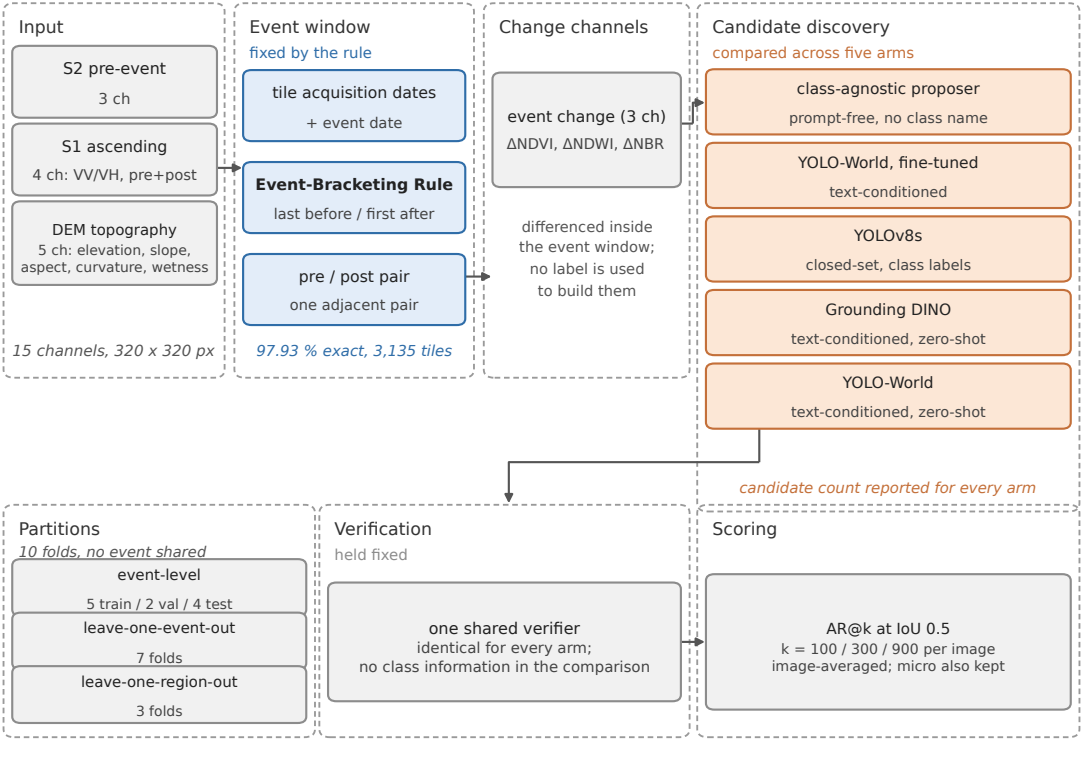
Input
S2 pre-event
3 ch
S1 ascending
4 ch: VV/VH, pre+post
DEM topography
5 ch: elevation, slope, aspect, curvature, wetness
15 channels, 320 x 320 px
Event window
fixed by the rule
tile acquisition dates
+ event date
Event-Bracketing Rule
last before / first after
pre / post pair
one adjacent pair
97.93 % exact, 3,135 tiles
Change channels
event change (3 ch)
ΔNDVI, ΔNDWI, ΔNBR
differenced inside the event window; no label is used to build them
Candidate discovery
compared across five arms
class-agnostic proposer
prompt-free, no class name
YOLO-World, fine-tuned
text-conditioned
YOLOv8s
closed-set, class labels
Grounding DINO
text-conditioned, zero-shot
YOLO-World
text-conditioned, zero-shot
candidate count reported for every arm
Partitions
10 folds, no event shared
event-level
5 train / 2 val / 4 test
leave-one-event-out
7 folds
leave-one-region-out
3 folds
Verification
held fixed
one shared verifier
identical for every arm;
no class information in the comparison
Scoring
AR@k at IoU 0.5
k = 100 / 300 / 900 per image
image-averaged; micro also kept
fixed by the rule
compared across arms
shared and held fixed